COLIN NGUYEN

San Francisco Bay Area • cnguyen3944@gmail.com • linkedin.com/in/cnguyen25 • github.com/cnguyen75

AI Engineer building production LLM agents and the evaluation infrastructure that keeps them reliable in regulated fintech — agent design and orchestration, rigorous LLM evaluation, and coding-agent workflows, owned end-to-end.

EXPERIENCE

Athena Money *Nov 2025 – Present*

Founding AI Engineer *San Francisco Bay Area*

- Designed and built the team's LLM evaluation platform for multi-turn agents in regulated mortgage underwriting: three evaluator types (LLM-as-judge rubrics, per-turn quality scoring, outcome checkpoints) driven by an LLM user-simulator.
- Built field-level consensus evaluation — running an agent across N trials and computing per-field agreement with confidence-score calibration — plus statistical regression gating in tiered CI, with the eval driver on durable Temporal workflows for reproducible reruns.
- Shipped production underwriting agents on Google ADK and Temporal, built so that every action is validated in the agent's turn, fails closed, and is approved by a person before it takes effect.
- Migrated a legacy workflow onto a multi-turn agent pattern one request type at a time behind feature flags, with the old path as fallback, then retired the legacy system.
- Built scheduled coding-agent workflows that audit the product and open pull requests, with audit events and a merge boundary enforced by the dispatcher rather than the prompt.
- Built document-verification and extraction pipelines for regulated documents, failing closed on malformed fields.

Zeta Global *Jun 2025 – Aug 2025*

Data Science & Generative AI Intern *San Francisco*

- Engineered a LangGraph supervisor agent that interprets and routes tasks to sub-agents, cutting manual workflow setup time by 70%.
- Built a multi-vendor (OpenAI, Anthropic, Bedrock) orchestration platform on FastAPI with PostgreSQL checkpointing, plus an evaluation framework (accuracy, consistency, latency, cost).
- Co-invented a patent-pending adaptive feedback system ("Adaptive Neural Content Generation") pairing LLMs and neural networks in an RL loop; deployed agents via REST APIs on Docker and Kubernetes.

NationBuilder *Jan 2025 – Nov 2025*

AI Research Intern (UCSB capstone project through Jun 2025) *Remote*

- RAG pipeline (all-MiniLM-L6-v2 embeddings, FAISS) ranking document relevance at 93% relevance-classification accuracy.
- Capstone: scraped election data and built an agentic PDF-extraction system (GPT-4o) with strict JSON schemas.

SELECTED PROJECTS

Decision Gates — research with Prof. Eric Suess, CSU East Bay *2026 – Present*

- Pre-registered study of risk-controlled abstention under option-count shift; fine-tuned SmolLM2 decision models and four calibration regimes on a laptop: the guarantee fails from 5 to 150 options on a public model and on mine.

Efficient LLM Fine-Tuning — research with Prof. Eric Suess, CSU East Bay *2026*

- Five stacked memory-saving techniques raised SmolLM2's trainable step from 8,192 to 131,072 tokens in 71% of baseline memory; a hand-written block-wise 8-bit AdamW matched torchao's to the byte.

Pokémon TCG AI Battle Challenge — Kaggle *2026*

- Agent for a hidden-information card game on a live ranked ladder: scripted expert policy, determinized whole-turn beam search, and a value network trained by self-play.

LLM 20 Questions — Kaggle *2024*

- Two-model agent on 4-bit quantized LLMs (Transformers, BitsAndBytes): one asks and guesses, one answers.

SKILLS

Languages: Python, Go, SQL, Java, JavaScript, R, C

LLM / AI: multi-agent orchestration (LangGraph, Google ADK, Temporal), RAG, vector search (FAISS), embeddings, LLM evaluation / LLM-as-judge, MCP (Model Context Protocol), prompt engineering, LangSmith, OpenAI / Anthropic / Bedrock APIs

ML: PyTorch, fine-tuning / PEFT (LoRA, QLoRA), vLLM, scikit-learn, TensorFlow, Node2Vec, feature engineering, calibration

Infra / MLOps: AWS (Bedrock), Google Cloud, Docker, Kubernetes, FastAPI, PostgreSQL, CI/CD, observability, Spark, Databricks

EDUCATION

University of California, Santa Barbara *Sep 2022 – Jun 2025*

B.S. in Statistics & Data Science, Minor in Mathematics *Santa Barbara, CA*

Patent pending: "Adaptive Neural Content Generation" — U.S. Provisional App. No. 63/943,780, filed Dec 2025 (first-named inventor).